Telecommunication Engineering Centre
(Department of Telecommunications)
Khurshid Lal Bhawan, Janpath, New Delhi -110001
(Radio Division)

No. 8-1105/2021-R/TEC **Date: May 31, 2023**

Sub: Minutes of National Study Group- 5 Meeting held on 24.05.2023.

A meeting of National Study Group-5 (NSG-5): Terrestrial Services for ITU-R for study cycle 2019-23 was held on 24.05.2023 in online mode.

List of participants is attached as Annexure.

1. The meeting began with Welcome Address by Ms. Bhoomika Gaur, ADG (Radio), TEC and opening remarks of Shri. Avinash Agarwal, DDG (Radio) and coordinator of NSG-5 in which he appreciated the contributing members of NSG-5 and encouraged them to continue to make notable contributions to ITU-R WP 5D and ITU-R. This was followed by a brief introduction of the participants. Thereafter, Sh. L.D. Meghwal, Sr. DWA, WPC and Rapporteur (NSG-5) briefed about the agenda of the meeting and proceeded with the discussions.
2. Further, it was mentioned by Ms. Neha Upadhyay, Director (R-I) that Shri. R. R. Mittar, Sr. DDG & Head (TEC) and Chairman (NSG-5) could not attend this meeting due to parallel engagements. Hence, all the decisions of this meeting would serve as Recommendation and the final authority for approval/rejection of any proposal rests with Chairman.
3. A summary of the discussions is as under –

   i. **Contribution I: Proposal for development of new ITU-R Report on IMT-2020 Deployment Experiences**

   Contributors: TEC and M/s 5GIF

   This document pertaining to IMT 2020 Deployment Experiences was presented by Dr. Sendil Kumar, M/s 5GIF. He proposed that a new ITU-R Report for 'IMT-2020 Deployment Experiences' be developed by WP 5D during the study cycle of 2023-2027 to include information about the actual experience of IMT- 2020 networks deployments, spectrum allocation, user experiences, or any other specific regulatory or operational measures taken to enable IMT-2020 deployments. The Report could also include aspects like spectrum utilisation, throughput delivered, usage scenarios realized etc in various environments such as dense urban, hotspots, indoor etc. A detailed workplan for the same was also provided in the report.

   NSG-5 Deliberations:
   It was suggested that the scope of the document may be elaborated to provide more clarification. Also, stakeholders, including M/s COAI, suggested that data about the total number of base stations along with spectrum bands that are in use can also be included as it will be helpful in knowing the density of IMT stations in an area. Further,

representatives from M/s GSMA and M/s Nokia supported the proposal and submitted that such a document will be of significance to a lot of markets, typically South Asian markets, where 5G services are yet to be rolled out as this document (proposed ITU-R Report) may provide data for best practices and better co-existence scenarios. Considering above views, ***the contribution was recommended by Rapporteur NSG-5 for further consideration by Chairman NSG-5.***

ii. **Contribution II: Proposal towards the Draft new Recommendation ITU-R M.[IMT.FRAMEWORK FOR 2030 AND BEYOND]**

Contributor: TEC

The contribution was presented by Ms. Bhoomika Gaur, ADG (Radio), TEC. The document highlighted India's recently launched 'Bharat 6G Vision' document and proposed that the ITU-R Draft new Recommendation on Framework for IMT 2030 and beyond may be finalized with- inclusion of India's 6G Vision Document on ITU-R IMT-2030 webpage and inclusion of its select IMT elements in the draft Recommendation.

NSG-5 Deliberations:
M/s COAI, M/s Tejas Networks and M/s 5GIF supported the proposal and suggested minor editorial changes in the document. M/s Nokia suggested that interconnectivity of terrestrial and non-terrestrial networks should be well defined. M/s Tejas also suggested that this contribution should be sent with the one proposed by them as a single joint contribution. Based on discussions, ***the contribution was recommended by Rapporteur NSG-5 for further consideration by Chairman NSG-5.***

iii. **Contribution III: Approval of the Recommendation on the IMT-2030 FRAMEWORK**

Contributor: M/s 5GIF

This document was presented by Dr. Sendil Kumar, M/s 5GIF. The document suggests that the proposed time plan for the national vision on 6G (Bharat 6G Vision) is very much in alignment with ITU-R WP 5D workplan. In order to be certain of the spectrum ranges in which 6G technologies will be developed, he proposed that the work of development of ITU-R Draft new Recommendation on Framework for IMT 2030 and Beyond be completed in 44th meeting of WP5D.

NSG-5 Deliberations:
It was mentioned by TEC members and NSG-5 Rapporteur that while India would work for completion of this WP5D document, there are still multiple requirements, submitted by Indian administration earlier, which need to be included in the document. Hence, India would refrain from sending an explicit statement or support for completion of the document at this 44th meeting of WP5D. ***Hence, this contribution was not recommended by Rapporteur for submission to ITU-R WP5D. This proposition was to be further considered by Chairman NSG-5.***

iv. **Contribution IV: Proposal for further update to [Preliminary Draft New Recommendation or the Working Document towards a Preliminary Draft New Recommendation] ITU-R M.[IMT.FRAMEWORK FOR 2030 AND BEYOND**

Contributors: M/s Tejas Networks, IIT Hyderabad, M/s Reliance-JIO and et.al.

Mr. Jishnu from M/s Tejas Networks presented this contribution pertaining to CG Chair output for further update in the following sections:

- Section 2.2- User and Application Trends
- Section 3- Usage Scenarios for IMT for 2030 and Beyond
- Section 4- Capabilities for IMT for 2030 and Beyond

He proposed that "Energy Efficiency" either be kept as a separate capability or as a sub-section of the "Sustainability" capability. He added that specific requirements for usage scenarios of ubiquitous connectivity and capability be met, in order for the document to be upgraded from a working document to a PDNR. He further recommended some editorial changes and also suggested the removal of square brackets for certain words and target ranges. However, other members including representatives from M/s Nokia, M/s Reliance JIO and M/s GSMA emphasized that some editorial changes may be incorporated in order to provide better clarity. M/s 5GIF suggested that the document should also encompass fallback strategy. Additionally, many members suggested that use of harsh/obstructionist language such as '*unless the above requirements of a separate Usage Scenario for Ubiquitous communication and Capabilities are considered the document is not mature enough to be upgraded from Working Document to PDNR*' may be avoided and the proposal be re-worded in a more diplomatically appropriate language.

NSG-5 Deliberations:
This contribution was recommended for submission to ITU-R with minor editorial changes and inclusion of supporting text, later to be embedded in the document. Also, NSG-5 recommended that use of harsh/obstructionist language, such as '*unless the above requirements of a separate Usage Scenario for Ubiquitous communication and Capabilities are considered the document is not mature enough to be upgraded from Working Document to PDNR*' should be avoided and the contribution should be re-submitted with more diplomatically appropriate text while asserting our stand. T***he contribution was recommended by Rapporteur NSG-5, after incorporating NSG-5 suggestions, for further consideration by Chairman NSG-5.***

v. **Contribution V: Supporting material for Working Document and for the draft CPM text for WRC-23 AGENDA ITEM 1.1**

Contributor: WPC

This document pertaining to technical contributions related to WRC-23 AGENDA ITEM 1.1 was presented by Sh. M.P.S. Alawa, Sr. DWA, WPC Wing. The document contains various pfd limits (existing value = -155 dB/(W/(m^2 per MHz))) for protection of Aero Mobile Station (AMS) and Maritime Mobile Station (MMS). He reflected that different studies carried out to calculate the pfd limits gave different results for both AMS and MMS. He further added that having this contribution will be advantageous in WRC as this document is an extension towards studies that have already been done. This proposal was supported by M/s IAFI.

NSG-5 Deliberations:
The contribution was supported by M/s IAFI and M/s Tejas Networks along with some minor editorial changes. It was mentioned by Rapporteur that WP5D currently does not have any workplan on Agenda Item 1.1 and the CPM text has already been developed by 5D and submitted to CPM. So, it is uncertain how this contribution will be received by WP5D or CPM. On this, proponent clarified that this text/proposal may not be accepted by CPM but submitting the same in WP5D will put India's contribution on record and it will help in defending India's position in WRC. M/s IAFI also supported the strategy of submitting this contribution to bring on record the proposed text (pfd limits). ***The contribution was recommended by Rapporteur NSG-5 for further consideration by Chairman NSG-5.***

vi. **Contribution VI: Amendment on Working Document towards a Preliminary Draft New Report ITU-R M.[IMT.AAS]**

Contributor: WPC

This contribution was presented by Sh. M.P.S. Alawa, Sr. DWA, WPC Wing. He elaborated that the report includes editorial changes in comparison to its previous versions. This document reflects on the nature of Advanced Antenna System or Active Antenna System (AAS) in 3GPP. He added that on simulating the equations mentioned in Recommendation ITU-R M.2101 in various form in MS Excel and MATLAB, it was noted that certain technical parameters were required to be limited, beyond which different mitigation technique will have to be adopted. This contribution intends to bring more clarity about AAS.

NSG-5 Deliberations:
It was mentioned that this contribution was earlier submitted from India to WP5D (5D/320) and is currently a part of square bracketed report. Hence, Representative from M/s Samsung added that such simulation studies proposed in the document, are required to be done by industry also for long term plan and hence small groups can be formed for more discussions. After these deliberations, ***this contribution was recommended by Rapporteur NSG-5 for further consideration of Chairman NSG-5.***

4. After much deliberations and discussions, it was suggested that supporting texts are required for some clauses which are to be provided by M/s Tejas Networks and M/s Reliance JIO, to make them more acceptable. Further, it was debated whether the recommended contributions on Framework for IMT 2030 and Beyond (from TEC and M/s Tejas and et.al.) should be submitted as single joint contribution or as separate contributions. On this, Rapporteur NSG-5 recommended that the source documents for these two contributions are different: (i) CG output document for sections 2.2, 3, 4 and (ii) WP5D Chairman Report for rest of the sections. Hence from regulatory point of view, there have to be two documents to bring clarity in the contribution. However, it was emphasized that the proposals in both the contributions should be aligned so as to maintain a single position of Indian administration in the two documents.

5. Director (Radio), TEC concluded the NSG-5 meeting with thanks to the chair and all participating members.

6. **Final Decision of Chairman NSG-5 on Contributions submitted to NSG-5:**
The meeting proceedings and recommendations of Rapporteur NSG-5 were then examined by Shri. R. R. Mittar, Sr. DDG & Head (TEC) and Chairman NSG-5. Based on Chairman's decision, the recommendations of NSG-5, on each contribution, are as follows:

i. **Proposal for development of new ITU-R Report on IMT-2020 Deployment Experiences**

This contribution was re-submitted by M/s 5GIF after making minor modifications in the document based on NSG-5 discussions. Further some comments were received via email from Sh. Jitendra Singh, M/s Qualcomm suggesting change of name of the report on the basis of past deliberations of WP5D. Hence, the new name of the proposed new ITU-R Report has been modified to: 'ITU-R REPORT ON NATIONAL APPROACHES FOR IMT-2020 DEPLOYMENT'.

This modified contribution from TEC and M/s 5GIF titled:
Proposal for development of new ITU-R Report on National approaches for IMT-2020 deployment
has been approved by Chairman NSG-5 for submission to ITU-R WP5D 44th meeting. Also, the same may be sent to WPC for further processing and onward submission to ITU-R.

ii. **Proposal towards the Draft new Recommendation ITU-R M.[IMT.FRAMEWORK FOR 2030 AND BEYOND]**
On the three contributions submitted in NSG-5 on ITU-R Draft Recommendation on Framework for IMT 2030 and Beyond, following has been decided has per discussions in NSG-5 meeting and remarks of Rapporteur NSG-5-

a. The contribution from M/s 5GIF has been dropped by Chairman NSG-5, as recommended by Rapporteur, NSG-5.

b. The contribution from M/s Tejas and et.al., after incorporating NSG-5 suggestions and rewording the proposal in appropriate language, has been approved by Chairman NSG-5 for submission to ITU-R WP5D 44th meeting. Also, the same may be sent to WPC for further processing and onward submission to ITU-R.

c. Regarding contribution from TEC, it was mentioned that the contribution is good and appreciable. Further, it has been decided that a single contribution on Framework for IMT-2030 and Beyond should be submitted from India so as to maintain coherence and uniformity in Indian proposal(s). Hence, for submission in the 44th meeting of WP5D, the select IMT elements from Bharat 6G Vision document, as proposed in TEC's contribution, have been merged in the contribution of M/s Tejas and et.al.
Further the proposal of endorsing India's 6G Vision document on IMT 2030 webpage is currently postponed and the same may be considered for submission in subsequent ITU-R/WP5D meetings.

d. Hence, a single contribution on Framework for IMT 2030 and Beyond has been finalized by merging contributions from TEC and M/s Tejas and et.al and the same has been approved by Chairman NSG-5 for submission to ITU-R WP5D 44th meeting. It may be sent to WPC for further processing and onward submission to ITU-R.

iii. Supporting material for Working Document and for the draft CPM text for WRC-23 AGENDA ITEM 1.1

This contribution has been approved by Chairman NSG-5 for submission to ITU-R WP5D 44th meeting. Also, the same may be sent to WPC for further processing and onward submission to ITU-R.

iv. Amendment on Working Document towards a Preliminary Draft New Report ITU-R M.[IMT.AAS]

This contribution has been approved by Chairman NSG-5 for submission to ITU-R WP5D 44th meeting. Also, the same may be sent to WPC for further processing and onward submission to ITU-R.

Bhoomika
31/05/2023
(Bhoomika Gaur)
ADG (Radio), TEC

ANNEXURE

List of participants of the NSG-5 Meeting held on 24.05.2023.

1. Sh. Avinash Agarwal, DDG (Radio), TEC
2. Sh. Arun Agarwal, DDG, J&K LSA
3. Sh. Ashish Tayal, GM, M/s BSNL
4. Sh. L.D. Meghwal, Sr. DWA, WPC Wing and Rapporteur (NSG-5)
5. Sh. M.P.S. Alawa, Sr. DWA, WPC Wing
6. Ms. Neha Upadhyay, Director (Radio), TEC
7. Sh. V. K. Roy, Director (MT-I), TEC
8. Ms. Bhoomika Gaur, ADG (Radio), TEC
9. Mr. Anshul Kumar Gupta, AD (Radio), TEC
10. Mr. Ankit Adjariya, AD, DCPW, MHA
11. Assistant Director, DCPW, MHA
12. Mr. Piyush Goyal, AD, DCPW, MHA
13. Mr. Kumaramohan, DoS/ISRO
14. Ms. Pamela Kumar, TSDSI
15. Prof. Chandra R. Murthy, IISc
16. Prof. Kiran Kuchi, IIT Hyderabad
17. Dr. Vinosh James, M/s Qualcomm and 5GIF
18. Dr. Sendil Kumar, M/s 5GIF
19. Mr. Abhijit Panicker, M/s COAI
20. Mr. Vikram Tiwathia, M/s COAI
21. Mr. Ashish Garg, M/s GSMA
22. Mr. Bharat Bhatia, M/s IAFI
23. Mr. Deepak Yadav, M/s Nokia
24. Mr. Dileep Lakhera, M/s Bharti Airtel
25. Mr. Diwakar, M/s Samsung
26. Mr. Jishnu, M/s Tejas Networks
27. Mr. Satish Jamadagni, M/s Reliance JIO
28. Mr. Vinay Shrivastava, M/s Reliance JIO
29. Mr. Junaid
30. Mr. J. Siddiquee
31. Mr. R. P.